DEPARTMENT OF ENGINEERING

POLITECNICO DI TORINO

Master's Degree in Environmental Engineering

Course: Renewable Energy Resource

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Preface

This report presents the development of an integrated energy and waste management system for the Gardaland Resort, conceived according to the principles of circular economy and sustainable resource management.

The primary objective of the project is to investigate the technical feasibility of transforming the resort into a largely energy self-sufficient facility through the integration of renewable energy technologies and waste-to-energy processes. To achieve this goal, several energy generation scenarios were evaluated, including photovoltaic systems, wind energy, and anaerobic digestion of locally available organic residues and waste streams.

The project was developed as part of the academic activities of the course attended during the Academic Year 2025/2026. Consequently, all analyses, assumptions, design methodologies, and sizing procedures were carried out in accordance with the approaches, design criteria, and engineering methodologies presented throughout the course.

The results presented in this report should therefore be interpreted as a technical-academic exercise aimed at applying theoretical and practical engineering concepts to a realistic case study, rather than as a final executive design.

1 Introduction: Gardaland's current energy mix

Gardaland Resort, located in the municipal area of Castelnuovo del Garda in Northern Italy, stands as one of the most visited and prominent entertainment hubs in Southern Europe. Situated near the shores of Lake Garda, the resort's surrounding territory is predominantly agricultural, characterized by a significant presence of vineyards and olive groves. This specific geographical and topographical context not only influences the local microclimate but also dictates the availability of natural resources for energy generation. The complex encompasses the main Theme Park, the Legoland Park, the Sea Life aquarium, and three themed hotels. Accommodating approximately 3,000,000 visitors annually alongside 250,000 hotel overnight stays, this vast ecosystem requires a continuous and highly reliable electricity supply. Critical inductive loads arise from roller coasters, large-scale water-pumping systems, extensive climate control infrastructure, and lighting systems, as well as the 24/7 operations of the resort's hospitality and restaurant facilities

The resort's operations are characterized by strong seasonal variability, with energy demand peaking during the spring and summer months when visitor attendance is highest. The aggregate annual electricity demand is estimated at 23.0 GWh/year, with the theme park attractions alone accounting for approximately 14.4 GWh/year. Currently, a 3 MWp photovoltaic canopy system installed over the parking areas, developed in partnership with ENGIE, contributes approximately 5.75 GWh/year (25% of total demand). Despite this infrastructural asset, the remaining 75% of the required electricity is procured from the Italian national grid. This heavy reliance exposes the resort to significant energy price volatility and generates a substantial indirect carbon footprint (approximately 15,525 tonnes of CO_2 per year based on the average Italian grid emission factor of 450 gCO_2/kWh).

To address this vulnerability and align with circular economy principles, this report investigates the technical and economic feasibility of transforming the resort into a largely energy self-sufficient facility. To achieve this goal, four distinct renewable energy configurations are modeled and evaluated: a 100% solar energy expansion supported by battery storage (Scenario 1), a 100% off-site wind energy integration leveraging the Rivoli Veronese corridor via a virtual Power Purchase Agreement (Scenario 2), a fully integrated mix incorporating wind, solar, and a newly designed anaerobic digestion plant for the treatment of local organic waste (Scenario 3), and a balanced 50% solar and 50% wind hybrid system (Scenario 4). By comparatively assessing these diverse generation pathways, this study aims to identify the optimal infrastructural and financial strategy to fully decarbonize the resort's energy profile.

2 Wind energy

This section explores the feasibility and strategic planning of wind power integration to support the project's energy demands. Since the local wind resources at the primary site are insufficient, the focus shifts to identifying an optimal off-site generation node. The following subsections detail the selection of a highly productive wind corridor and provide a comprehensive statistical analysis of its historical wind data to validate the site's energy generation potential.

2.1 The Rivoli Veronese wind corridor

The immediate surroundings of Castelnuovo del Garda are characterised by low macro-zonal wind speeds, making any on-site wind generation economically and technically unviable. To overcome this geographic constraint, the project targets the existing industrial wind zone of Monte Mesa in Rivoli Veronese (VR), situated approximately 15–18 km north of the Gardaland Resort.

This specific site is one of the most productive onshore wind corridors in Northern Italy, owing to its unique topography. Positioned at the confluence of the Adige Valley and the pre-Alpine terrain, the location acts as a natural aerodynamic nozzle. As large air masses descend from the Alpine region toward the Po Valley, they are compressed through the narrow valley walls, generating a permanent micro-climatic acceleration known as the Venturi effect. This topographical phenomenon ensures consistent, high-velocity directional wind streams that are ideally suited to commercial wind energy exploitation.

An additional strategic advantage of this site is that wind infrastructure is already an established part of the landscape (4 existing REpower MM92 turbines of 2.05 MW each are already operational), which significantly reduces permitting complexity and community opposition for a capacity expansion.

2.2 Wind data analysis

Wind data for the site was sourced from the ERA5 reanalysis dataset provided by Copernicus (European Centre for Medium-Range Weather Forecasts) for the period from 7 May 2016 to 7 May 2026 (10-year dataset of 87,672 hourly observations). Data was available as u and v wind components at a height of 100 metres. The resultant wind speed was calculated using:

$$v_{100} = \sqrt{u_{100}^2 + v_{100}^2} \tag{2.1}$$

Since the selected turbine model has a hub height of 90 m, the wind speed was corrected to the hub height using the logarithmic wind profile law:

$$v_{90} = v_{100} \cdot \frac{\ln\left(\frac{90}{z_0}\right)}{\ln\left(\frac{100}{z_0}\right)}, \qquad \text{where } z_0 = 0.2 \text{ m} \tag{2.2}$$

The mean annual wind speed at hub height (90 m) was calculated at 5.93 m/s, confirming the site's strong wind resource. The analysis of the hourly-monthly distribution reveals a clear seasonal and diurnal pattern: wind speeds tend to peak during late evening, night, and early morning hours, and are generally higher in winter and spring months (particularly March). This night-peaking profile provides a critical strategic complementarity with the existing solar PV system, whose output is confined to daylight hours. This trend is represented in Figure 2.1, which shows data sourced from Global Wind Atlas.

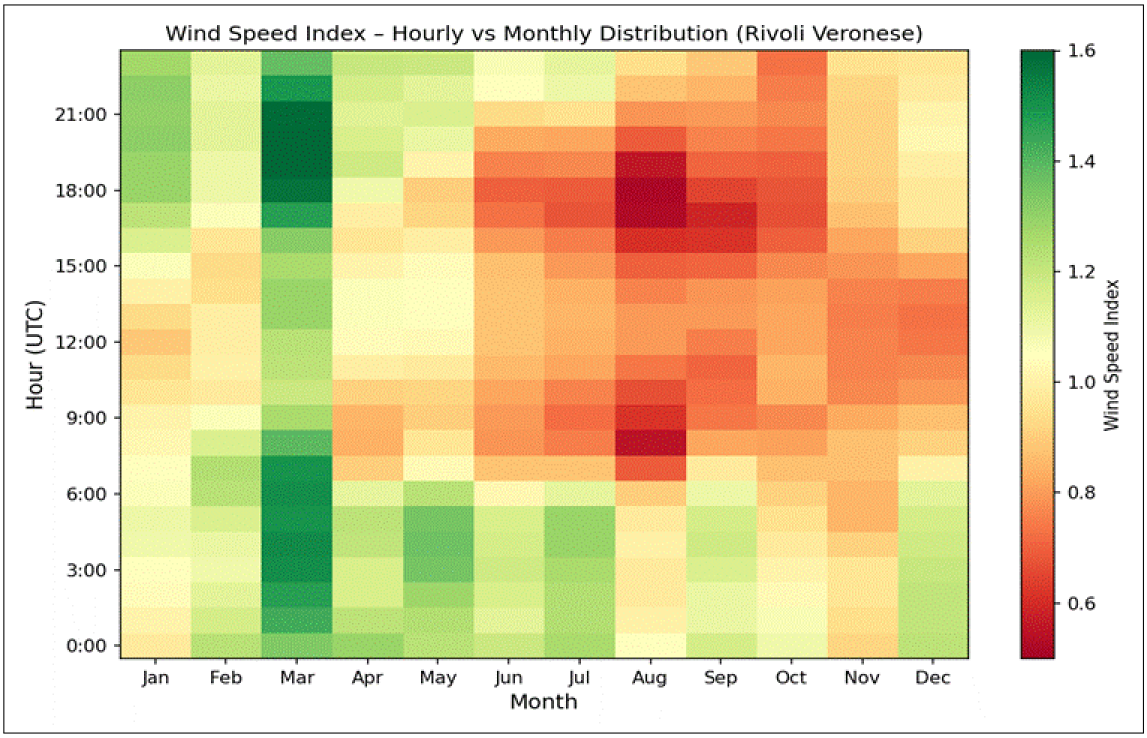

Figure 2.1: Wind speed index – Hourly vs monthly distribution (Rivoli Veronese, 2016–2026). Warmer colours indicate above-average wind speeds, confirming strong nocturnal and winter-season resource availability.

To convert raw wind speed time series into statistical production availability, the frequency distribution of wind speeds was computed by binning all 87,672 hourly observations into 0.5 m/s class intervals. The resulting Probability Density Function (PDF) was modelled, confirming a distribution broadly consistent with a Weibull function, with a modal wind speed approximately in the range of 6–8 m/s.

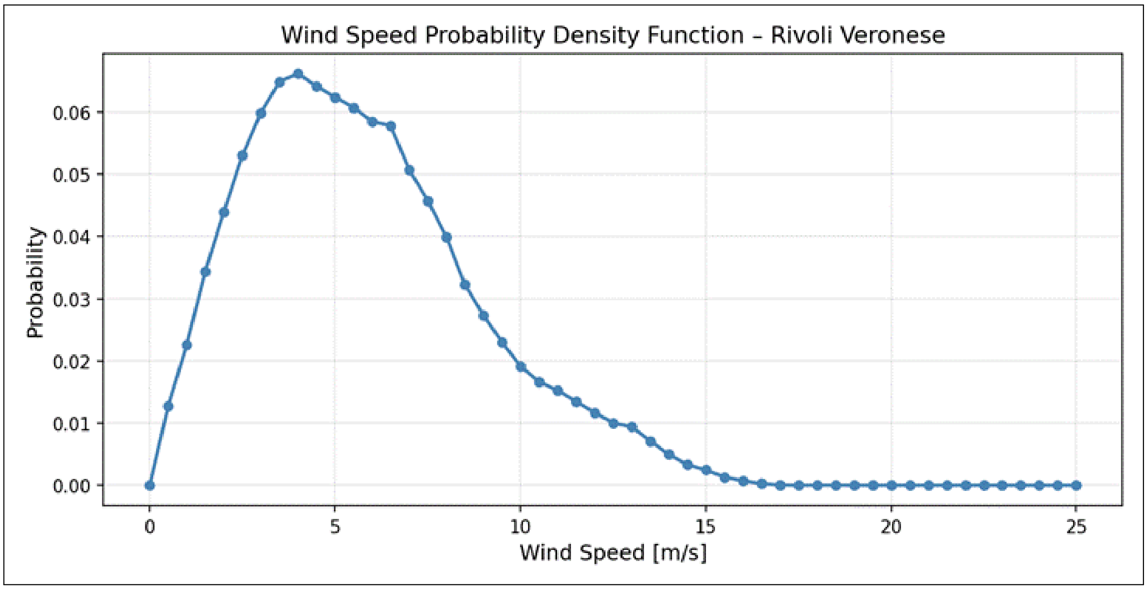

Figure 2.2: Wind speed Probability Density Function – Rivoli Veronese (2016–2026). The distribution shows a peak probability near 7 m/s, well within the operational range of the selected turbine model.

3 Solar Energy

This section of the report focuses on the potential implementation of solar energy technologies within the Gardaland amusement park and its associated resort facilities. As one of the largest and most visited amusement parks in Italy, Gardaland requires substantial amounts of electricity to support attractions, lighting systems, restaurants, hotels, air conditioning, and other services operating throughout the day. These characteristics make the park an ideal candidate for the integration of photovoltaic systems.

As confirmed by the renewable energy project already developed in collaboration with ENGIE Italia and Gardaland, the geographical location of Gardaland, in Northern Italy near Lake Garda, offers favorable solar irradiation conditions for the production of solar electricity, particularly during the spring and summer seasons when the park experiences its highest visitor attendance and energy demand. The installation of solar panels could therefore contribute significantly to reducing grid electricity consumption, lowering operational costs, and decreasing carbon dioxide emissions.

3.1 Load Analysis and Energy Demand Assessment

The design of the photovoltaic (PV) system began with a detailed assessment of the electricity demand of the study area. The available data included hourly, monthly, and annual electricity consumption profiles of the park and the surrounding hotels (fig. 3.1). The total annual electricity demand was estimated to be approximately 23 GWh.

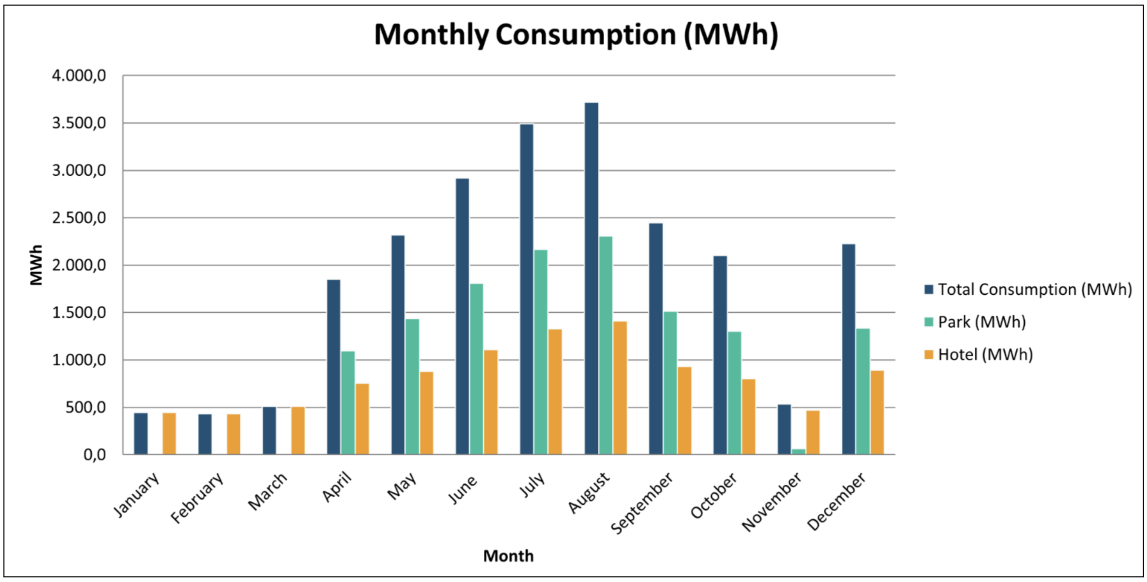

Figure 3.1: Monthly consumpion of Gardaland per hour

Based on these consumption data, the energy production target for the photovoltaic plant was determined. The main objective of the project was to design a PV system capable of generating enough electricity to cover the annual energy demand of the park and the nearby hotel facilities.

The already existant ENGIE project involves the construction of a large photovoltaic installation consisting of approximately 7,000 solar panels distributed over an area of about 14,000 m^2. The system has an installed capacity of 3 MW and is expected to generate approximately 3.6 GWh of renewable electricity per year. The produced energy is estimated to cover more than 25% of the total electricity demand of the park, including attractions, lighting systems, offices, and resort services. In addition to the energetic benefits, the project also contributes significantly to the reduction of environmental impacts, avoiding more than 800 tons of CO_2 emissions annually, corresponding to the environmental benefit of thousands of trees planted each year. Starting from the data provided by the ENGIE project, our analysis further evaluates how additional photovoltaic installations could increase the percentage of self-produced renewable electricity and contribute to making Gardaland an even more sustainable and energy-efficient entertainment facility. As a first design step, the required annual energy production was therefore set equal to the total annual electricity consumption minus the amount of energy produced by the ENGIE project. This target value served as the basis for the subsequent sizing of the photovoltaic system and the estimation of the installed capacity required to meet the demand.

3.2 Solar Irradiance

To begin our analysis, the annual solar irradiation data for Lake Garda area were obtained through PVGIS platform, one of the most widely used tools for photovoltaic energy assessment. The platform provides reliable meteorological and solar radiation data that can be used to estimate the energy production potential of photovoltaic systems in a specific geographical area.
Using coordinates corresponding to the Gardaland area near Lake Garda, the hourly solar irradiance values over an entire year were extracted. The resulting dataset, shown in Figure 3.2, represents the variation of solar radiation throughout the year expressed in W/m^2 as a function of time. This value corresponds to the optimal tilt angle of the panel, around 17, as provided by the European Photovoltaic Geographical Information System. It represents the inclination angle that maximizes energy production.

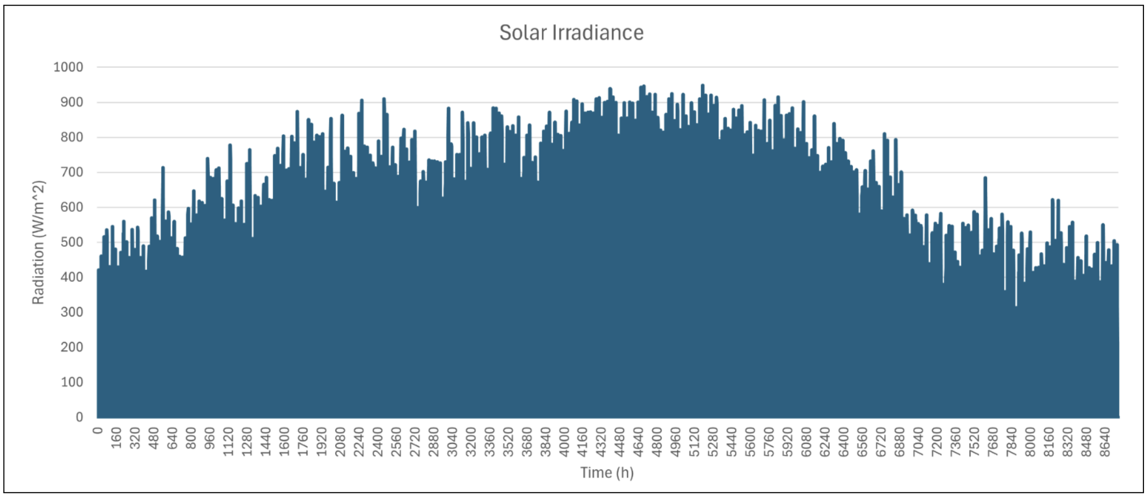

Figure 3.2: Solar Irradiance in Lake Garda's area.

After analyzing the energy consumption of both the amusement park and the associated resort facilities, it was possible to estimate the required size of the photovoltaic system in order to satisfy a significant portion of Gardaland's electricity demand. The sizing process was carried out by comparing the annual energy consumption of the entire site with the expected photovoltaic energy production obtained from the solar irradiance data provided by the PVGIS platform. Considering a crystalline silicon photovoltaic panel with an efficiency of 23% and estimated system losses equal to 10%, the corresponding energy production for the Gardaland area was calculated using the previously obtained solar irradiance data. The resulting annual energy per square meter as a cumulative curve output profile is reported in Figure 3.3

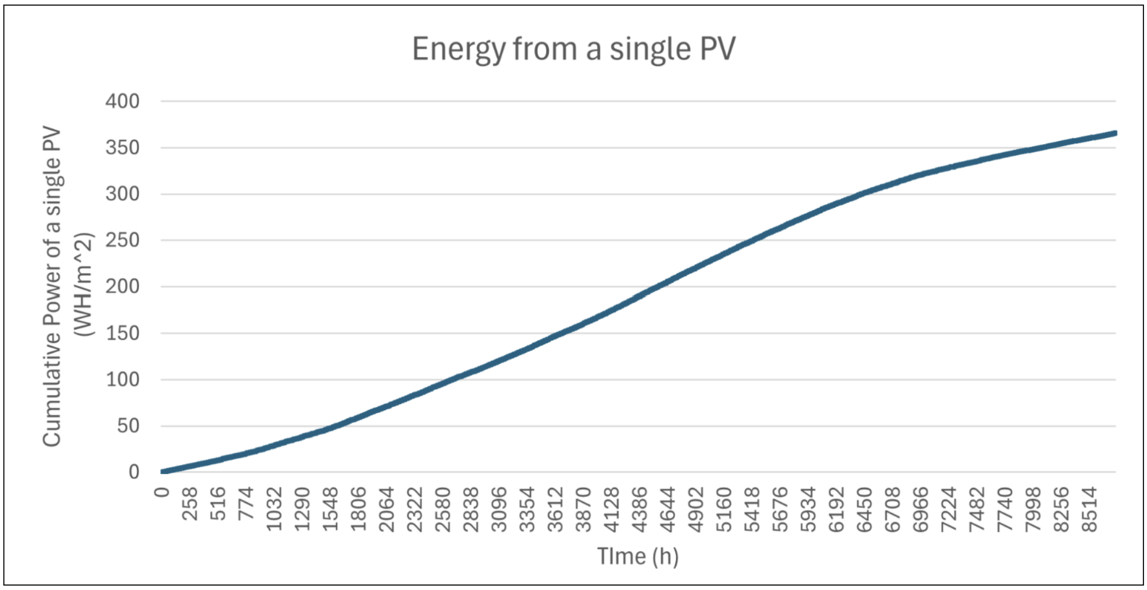

Figure 3.3: Cumulative energy from a single PV.

After estimating the annual energy production provided by one square meter of photovoltaic panels, it was possible to determine the total surface area required to satisfy the cumulative annual electricity demand required according to different scenario chosen to cover the total energy required by the park.

4 Anaerobic Digestion

This technical report focuses on the characterization of biomass feedstocks and the evaluation of the fundamental design parameters required for the preliminary sizing of an anaerobic digester serving the Gardaland amusement park and the surrounding municipal area of Castelnuovo del Garda.

The proposed facility aims to support the sustainable management of organic waste generated by a highly frequented tourist hub by integrating it with municipal organic fractions and agricultural residues available in the surrounding territory. This approach is fully consistent with circular economy principles, transforming waste streams into valuable resources through the production of biogas (or biomethane) and, potentially, digestate for agricultural soil amendment.

The design of the anaerobic digester is based on an accurate quantitative and qualitative characterization of the available feedstocks. The stability of the biological process and the methane yield strongly depend on the composition of the feed mixture (*feedstock mix*), particularly on its Dry Matter (DM) and Volatile Solids (VS) content. Therefore, the following sections analyze and quantify the different biomass sources considered for the plant, including organic waste generated within the Gardaland resort, municipal organic waste, urban green maintenance residues, and agricultural by-products.

4.1 Feedstock Characterization

For the calculation of the reactor volume and the Organic Loading Rate (OLR), the available feedstocks were grouped into four main categories according to their origin and physico-chemical characteristics. All quantities are expressed on an annual basis.

4.1.1 Organic Waste Generation within the Gardaland Resort

The annual organic waste production associated with the Gardaland resort was estimated by disaggregating the contributions from three primary user categories: park visitors, staff members, and hotel guests.

- Park visitors: assuming an average annual attendance of 3,000,000 visitors and a specific food waste generation rate of 0.165 kg per visitor, the estimated production is approximately 495 t/year.
- Staff: considering 1,200 full-time equivalent employees working an estimated 230 days per year, the staff canteen generates approximately 27.6 t/year of organic waste.
- Hotel guests: based on approximately 250,000 annual overnight stays and a specific waste generation rate of 0.5 kg per stay, the estimated contribution is 125 t/year.

The aggregate organic waste production from the resort is therefore estimated at 648 t/year. The resulting feedstock is characterized by a Dry Matter (DM) content of 15% and a Volatile Solids (VS) fraction equal to 90% of the DM.

4.1.2 Municipal Organic Waste and Green Maintenance Residues

To complement the waste streams generated within the resort, additional biomass sources originating from municipal services in Castelnuovo del Garda were considered. Figure 4.1 illustrates the distribution of the various collected municipal waste fractions in the study area.

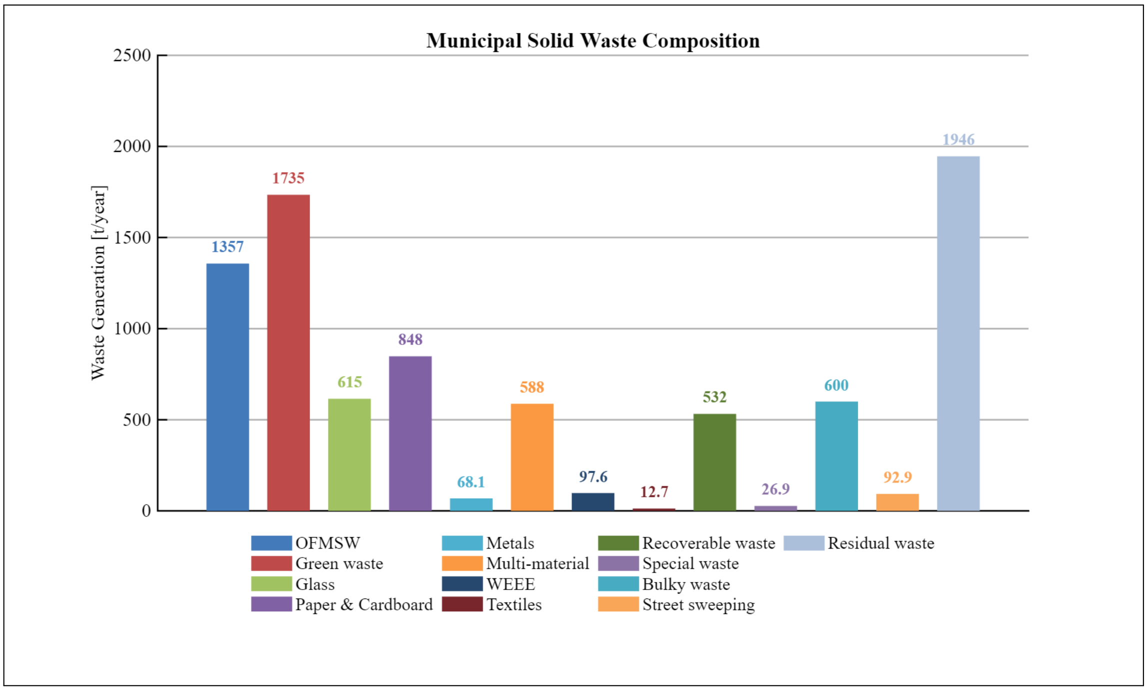

Figure 4.1: Distribution of separately collected municipal solid waste fractions. Data source: ISPRA Veneto.

These municipal sources are primarily divided into two main categories. The first, urban greenery residues, consists of pruning and grass-cutting waste generated during the maintenance of public areas. Representing the largest contribution from the municipal sector, this stream provides an estimated 1,734.86 t/year. The material features a dry matter (DM) content of 40%, with volatile solids (VS) accounting for 85% of the DM; however, its availability remains highly seasonal, typically exhibiting production peaks during the spring and autumn months. The second category, municipal organic waste (OFMSW), comprises the organic fraction of municipal solid waste gathered through separate collection schemes. By applying a suitability factor of 50% to the total available quantity, the effectively recoverable amount for this stream is estimated at 678.62 t/year. This specific feedstock is characterized by a DM content of 20% and a VS content equivalent to 85% of the DM.

4.1.3 Agricultural By-products and Residues

The surrounding territory is predominantly agricultural and therefore offers a significant and stable source of biomass suitable for anaerobic digestion. The assessment of the available feedstock was performed by combining the Utilized Agricultural Area (UAA), classified according to official ISTAT categories, with crop-specific residue generation factors expressed in tons of fresh matter per hectare.

The resulting theoretical biomass potential was subsequently adjusted through the application of availability coefficients to estimate the technically recoverable fraction. These factors account for agronomic and environmental constraints, including the need to retain part of the residues on the field to preserve Soil Organic Carbon (SOC), limit erosion phenomena, and ensure sustainable agricultural practices, as well as operational limitations associated with harvesting activities. Residue generation rates and availability coefficients were derived from recognized national and European technical references, including ENEA, CRPA, ITABIA, and JRC studies.

The final estimate of recoverable agricultural by-products and residues is reported in Figure 4.2.

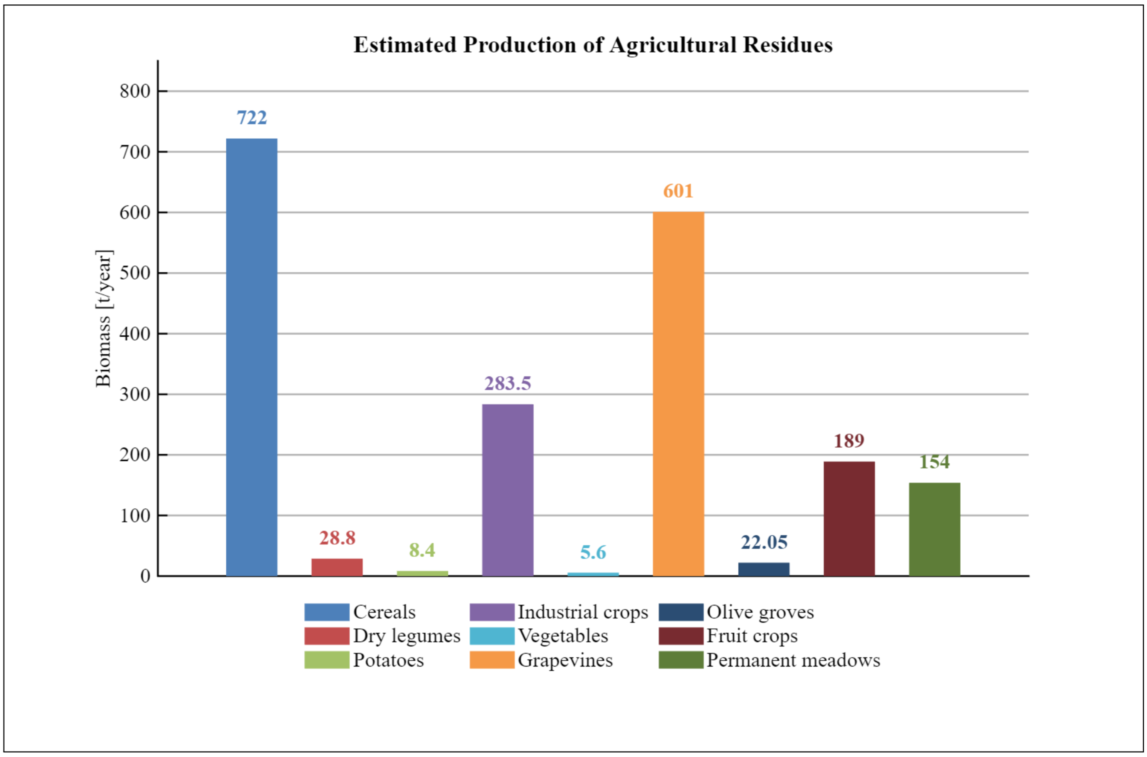

Figure 4.2: Estimated production of agricultural by-products and residues. Elaborated from ISTAT cultivated area data and European guideline conversion factors.

Based on the available agricultural resources, two potential feedstock supply scenarios were identified. The first scenario, defined as a complete agricultural mix, includes residues from cereal cultivation (straw and stalks from 361 ha), legumes, industrial crops, and pruning residues from fruit orchards. The total recoverable biomass for this option is estimated at 2,014.35 t/year, yielding a substrate characterized by an average DM content of 57% and a VS content equivalent to 92% of the DM. Alternatively, a second scenario focusing exclusively on olive grove and vineyard residues was evaluated. Given the predominance of vineyards (601 ha) and olive groves (21 ha) within the study area, this approach considers only pruning residues and associated biomass from these specific crops. This would provide an estimated available feedstock of 1,565.5 t/year, with a DM content of 50% and a VS content equivalent to 94% of the DM. Ultimately, the complete agricultural mix scenario was selected for the project in order to maximize the total recoverable biomass and ensure a higher feedstock availability for the process.

4.1.4 Summary of the Feedstock Mix

For the mass balance calculations and the estimation of the Organic Loading Rate (OLR), the complete agricultural mix scenario was selected as the reference case. The resulting feedstock composition is summarized in Table 4.1.

Table 4.1: Mass balance and physico-chemical characterization of the feedstock mixture.

Category	**Quantity (t/year)**	**TS (%)**	**VS (% of DM)**	**TS (t/year)**	**VS (t/year)**
Urban Greenery Residues	1,734.86	40	85	693.94	589.85
OFMSW	678.62	20	85	135.72	115.36
Gardaland Organic Waste	648.00	15	90	97.20	87.48
Agricultural Residues	2,014.35	57	92	1,148.18	1,056.33
TOTAL	**5,075.83**	**40**	**90**	**2,075.04**	**1,849.02**

The resulting feedstock mix amounts to approximately 5,076 t/year of raw biomass, corresponding to about 1,849 t/year of volatile organic matter. Furthermore, an analysis of these aggregate data—specifically the overall Total Solids (TS) content of 40%—clearly indicates that dry anaerobic digestion is the most appropriate technological choice. Achieving the lower solids concentrations required for a wet digestion system would necessitate a massive liquid addition, rendering the dilution process operationally and economically burdensome.

4.2 Pre-treatment Phases of the Biomass

To ensure the proper operation of the anaerobic digester, and consequently the effective progression of the anaerobic digestion stages, the biomass must undergo two primary pre-treatment steps, as outlined below: mechanical size reduction (shredding), aimed at achieving a uniform particle size distribution of the processed organic waste, and controlled dilution, intended to reach a specified total solids (TS) concentration within the mixture.

4.2.1 Shredding and Size Homogenization

The heterogeneity of the incoming feedstock—including the organic fraction of municipal solid waste (OFMSW), urban green residues, and lignocellulosic agricultural biomass—requires a mechanical pre-treatment stage aimed at size reduction and homogenization. This step is essential to increase the specific surface area available for microbial hydrolysis, enhance process kinetics, and prevent operational issues such as floating layers or sediment accumulation within the anaerobic digester.

The design of the shredding unit is based on the overall mass flow rate and the operational schedule of the plant, ensuring reliable performance under variable seasonal loading conditions. Particular attention is given to peak biomass supply periods, such as agricultural pruning seasons and high tourist influx periods. The plant is designed to treat an annual biomass input of approximately 5,075.83 t/year, characterized by a total solids (TS) content of 40% and a volatile solids (VS) fraction of 90% of TS. The operational schedule assumes 260 working days per year and 8 operating hours per day. Based on these assumptions, the theoretical mass flow rate is equal to 2.44 t/h. To ensure robust and reliable operation under non-ideal conditions, a safety factor of 1.20 is applied, resulting in a design throughput of 2.93 t/h. From an energy standpoint, the shredding process is characterized by a specific energy consumption of 12 kWh/t, leading to a useful power requirement of approximately 35.14 kW. Assuming an overall efficiency of 0.8, the theoretical installed power is estimated at 43.93 kW, which is rounded up to an installed power of 45 kW.

The main design assumptions and operating parameters used for the sizing of the shredding unit are summarized in Table 4.2.

Table 4.2: Design parameters of the shredding system.

Parameter	Value	Unit
Annual biomass throughput	5075.83	t/year
Operating days	260	day/year
Operating time	8	h/day
Design flow rate	2.93	t/h
Specific energy demand	12	kWh/t
Installed power	45	kW
Annual electricity consumption	60.91	MWh/year
Technology type	Double-shaft slow-speed shredder	-
Output particle size	< 40–50	mm

A slow-speed twin-shaft shredder has been selected as reference technology, suitable for mixed organic feedstocks and characterized by high torque, low rotational speed, and robust performance under heterogeneous loading conditions.

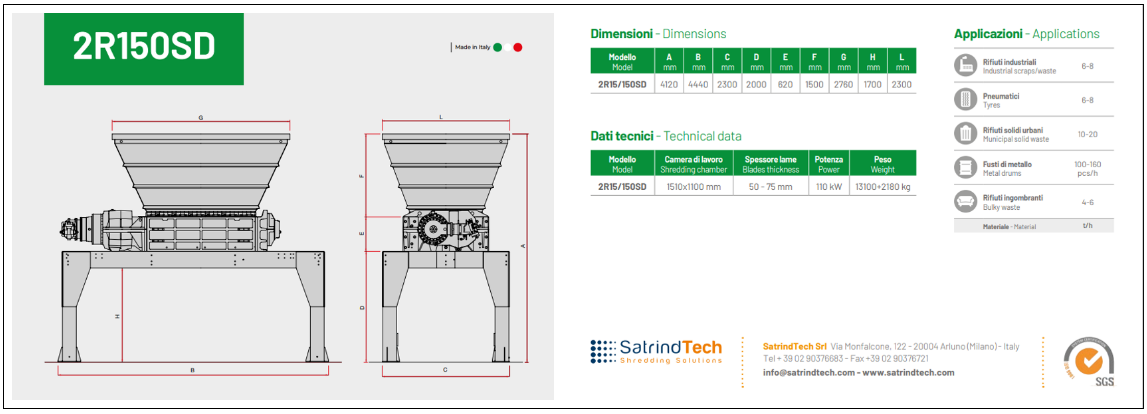

Figure 4.3: Technical specifications and application range of a twin-shaft industrial shredder (SatrindTech 2R series or equivalent).

Considering the calculated power demand of approximately 75 kW and the heterogeneous nature of the incoming biomass, several industrial shredders available on the market were evaluated as potential solutions. The most suitable models are reported in Table 4.3.

Table 4.3: Recommended Shredder Models

Recommended Model	Installed Power	Technical Justification
SatrindTech 2R 150/150	110 kW	Provides a substantial safety margin above the calculated power requirement (75 kW), ensuring reliable operation under variable loading conditions and during peak biomass supply periods. The machine is suitable for continuous-duty applications and offers high resistance to mechanical stress.
SatrindTech 2R 100/100	75 kW	Closely matches the estimated design power requirement and is suitable for standard operating conditions. Efficient operation can be achieved provided that the feed rate is properly controlled to avoid temporary overloads and excessive torque peaks.
Forrec TB1500	110 kW	Particularly suitable for heterogeneous and bulky biomass streams, including pruning residues and green waste. The higher installed power provides operational flexibility and ensures effective size reduction even under discontinuous feeding conditions.

4.2.2 Pre-mixing and Total Solids (TS) Adjustment in Dry Anaerobic Digestion

The incoming organic fraction is characterized by a Total Solids (TS) content of approximately 40%. This value is typical of high-solids substrates; therefore, extensive dilution, as required in wet digestion systems, is not necessary. However, a controlled pre-mixing stage remains essential to ensure the homogenization of the incoming feedstock, improve its pumpability and flowability, and adjust the Total Solids content to a value compatible with dry digestion operations. Specifically, the target Total Solids concentration is set to 25%, which falls within the typical operating range for dry anaerobic digestion systems (20–35% TS).

The plant is assumed to operate for 260 days per year, with 6 operating hours per day. Based on these assumptions, the mass balance of the pre-mixing stage is summarized in Table 4.4.

Table 4.4: Mass balance for the pre-mixing stage in dry anaerobic digestion

Parameter	Symbol	Value	Units
Operating days	d	260	d/year
Operating hours per day	h	6	h/day
Design feed flow rate	Q_{in}	3.00	t/h
Incoming TS content	TS_i	40	%
Target TS content	TS_t	25	%
Recycle flow rate	Q_{rec}	3.6	m^3/h
Total mixture flow rate	Q_{mix}	8.22	m^3/h
Pre-mixer volume	V	15	m^3
Specific power	P_s	1.4	kW/m^3
Number of mixers	n	4	–
Required power	P	21	kW
Nominal power	P_{nom}	18.5	kW

Based on the calculated power demand, a nominal installed power of 30 kW was selected. This choice is justified by the fact that the required power (approximately 22.5 kW) exceeds the nearest standard IEC motor size of 22 kW. The selected configuration therefore ensures adequate operational reliability, allows for a suitable service factor, and provides a margin for potential process fluctuations.

4.3 Anaerobic Digester Sizing

The anaerobic digestion unit was sized on the basis of the daily amount of substrate fed to the process. In order to ensure operational flexibility, process continuity, and the possibility of carrying out maintenance activities without completely shutting down the plant, a configuration consisting of two parallel horizontal plug-flow dry digesters was selected.

The incoming substrate mixture is delivered from the pre-mixing section at a flow rate of $Q_{\text{misc}} = 8.64\,\text{m}^3/\text{h}$. Since the upstream feeding system operates for only 8 hours per day, the corresponding daily feed volume supplied to the digesters is equal to:

$$Q_{\text{day}} = Q_{\text{misc}} \cdot h_{\text{pre}} \tag{4.1}$$

The feedstock entering the digestion process is characterized by a Total Solids (TS) content of 25% and a Volatile Solids (VS) fraction equal to 85% of the total solids. These values are representative of dry anaerobic digestion systems treating the Organic Fraction of Municipal Solid Waste (OFMSW).

The process is assumed to operate under mesophilic conditions, with a Hydraulic Retention Time (HRT) of 15 days. The required digestion volume was therefore calculated according to:

$$V = Q_{\text{day}} \cdot HRT \tag{4.2}$$

To provide additional operational reliability and accommodate possible fluctuations in feedstock composition and loading conditions, the digestion unit was designed with two identical reactors. Each reactor has an effective diameter of 6.0 m and an effective length of 24.0 m, corresponding to an effective volume of $678.58\,\text{m}^3$.

The total installed digestion capacity is therefore:

$$V_{\text{tot}} = n \cdot V_{\text{eff}} \tag{4.3}$$

The installed volume exceeds the minimum volume required by the retention time criterion, providing a design safety margin of approximately 31%. This reserve capacity improves process stability, accommodates fluctuations in organic loading, and contributes to safer reactor operation during transient conditions.

The main design parameters adopted for the anaerobic digestion system are summarized in Table 4.5.

Table 4.5: Main design parameters of the anaerobic digestion system

Parameter	Symbol	Value	Unit
Feedstock Characteristics			
Mixture Flow Rate	Q_{misc}	8.64	m^3/h
Operating Time	h_{pre}	8.00	h/day
Daily Feed Volume	Q_{day}	69.11	m^3/day
Total Solids	TS	25.00	%
Volatile Solids	VS	85.00	%
Process Design Parameters			
Hydraulic Retention Time	HRT	15.00	day
Required Digester Volume	V	1036.68	m^3
Number of Reactors	n	2	–
Reactor Geometry			
Tank Height	H	14.37	m
Tank Diameter	d	9.58	m
Effective Length	L_{eff}	24.00	m
Effective Diameter	d_{eff}	6.00	m
Effective Volume per Reactor	V_{eff}	678.58	m^3
Total Installed Volume	V_{tot}	1357.17	m^3

4.3.1 Organic Loading Rate (OLR) Verification

Table 4.6 shows the verification of the volumetric organic load, or Organic Loading Rate (OLR), for the adopted anaerobic digester configuration. Since it is a single-phase dry *plug-flow* reactor, the system is designed to operate with a high solids content. This allows it to handle significantly higher organic loads compared to traditional wet systems (CSTR). As can be seen from the calculations, the obtained design OLR value is 13.58 $kgVS/m^3day$. This result confirms the absolute validity of the volumetric sizing, as it falls perfectly within the optimal typical values reported in the literature for this specific type of plant (generally ranging between 12 and 15 $kgVS/m^3day$). This ensures an efficient conversion of the organic matter and the stability of the biological process, minimizing the risk of inhibition due to acid overload.

Parameter	**Value**	**Unit**
Qs	2.37	t/h
VS	85.00%	%
VSh	2.01	t/h
VSday	14.08	t/day
OLR	13.58	$kgVS/m^3day$

Table 4.6: Volumetric organic loading rate (OLR) verification

4.3.2 Mixing and Propulsion Power Requirements

In a single-phase dry *plug-flow* anaerobic digester, the high concentration of total solids significantly increases the viscosity of the fermenting biomass. Consequently, a robust mechanical mixing and propulsion system is crucial to ensure the continuous advancement of the substrate, maintain a homogeneous temperature distribution, and facilitate the release of the produced biogas while preventing the formation of dead zones.

For the design of this reactor, a specific power (P_s) of 15.00 W/m^3 was selected. This is a conservative and highly representative value for dry anaerobic digestion applications, where the rheological properties of the sludge demand substantial mechanical effort. Based on the individual digester volume, the theoretical required power (P_r) to mobilize the biomass was calculated at 15.55 kW per reactor.

To ensure operational reliability and to account for mechanical losses and high starting torque requirements, the total installed power (P_{inst}) for the dual-reactor system ($n = 2$) was sized at 37.00 kW, utilizing two standard 18.5 kW commercial electric motors. This selection provides a safety factor of approximately 1.19 (a 19% design margin) over the strictly required power, ensuring that the mixing system can handle temporary fluctuations in the organic matter's density without the risk of motor stalling or overload. The mixing power parameters are summarized in Table 4.7.

Parameter	Symbol	Value	U. M
Specific Power	P_s	15.00	W/m^3
Required Power (per reactor)	P_r	15.55	kW
Total Installed Power	P_{inst}	37.00	kW

Table 4.7: Mixing and propulsion power parameters for the anaerobic digestion system

4.3.3 Biogas Production and Energy Balance

The energy recovery potential of the anaerobic digestion plant was assessed on the basis of the annual amount of organic matter processed. The estimation of biogas production was performed using a specific biogas yield representative of dry anaerobic digestion systems treating the Organic Fraction of Municipal Solid Waste (OFMSW).

The annual biogas production was calculated from the quantity of biodegradable organic matter entering the process according to:

$$Q_{\text{biogas}} = M_{\text{org}} \cdot Y_{\text{biogas}} \tag{4.4}$$

where M_{org} is the annual mass of organic matter and Y_{biogas} is the specific biogas yield.

The produced biogas is converted into useful energy through a Combined Heat and Power (CHP) unit. The available thermal input power was estimated from the biogas flow rate and its Lower Heating Value (LHV), while the electrical and thermal outputs were determined according to the assumed CHP conversion efficiencies.

The thermal balance of the anaerobic digestion process was subsequently evaluated in order to verify the ability of the CHP system to maintain the reactors under mesophilic operating conditions. The heating demand was estimated from the mass flow rate of the incoming substrate, its specific heat capacity, and the required temperature increase:

$$P_{\text{heat}} = \dot{m}\, c_p\, \Delta T \tag{4.5}$$

Additional thermal losses associated with the reactor structure, piping network, and auxiliary equipment were considered in the overall heat demand.

The analysis demonstrates that the thermal energy recovered by the CHP unit is sufficient to satisfy the process heating requirements while maintaining a positive thermal balance. Consequently, a fraction of the recovered heat remains available for secondary uses, such as domestic hot water production, space heating, or integration with the thermal demand of adjacent facilities. This surplus contributes to improving the overall energy efficiency and economic performance of the plant.

The main parameters adopted for the biogas production assessment and energy balance are summarized in Table 4.8.

Table 4.8: Biogas production and energy balance parameters

Parameter	Symbol	Value	Unit
Biogas Production			
Organic Matter Content	M_{org}	1078.65	t/year
Specific Biogas Yield	Y_{biogas}	400.00	$\text{Nm}^3/\text{t}_{\text{org}}$
Annual Biogas Production	Q_{biogas}	431460.00	Nm^3/year
Average Biogas Flow Rate	q_{biogas}	53.93	Nm^3/h
Biogas Lower Heating Value	LHV	6.00	kWh/Nm^3
CHP Energy Recovery			
Available Input Power	P_{in}	323.60	kW
Electrical Power Output	P_{el}	129.44	kW
Thermal Power Output	P_{th}	161.80	kW
Thermal Balance			
Mass Flow Rate	$\dot{m}$	2879.65	kg/h
Specific Heat Capacity	c_p	4.18	kJ/(kg °C)
Temperature Increase	ΔT	30.00	°C
Process Heating Demand	P_{heat}	100.31	kW
Additional Heat Losses	P_{loss}	10.00	kW
Total Thermal Demand	P_{req}	110.31	kW
Available Thermal Surplus	P_{surplus}	51.49	kW

4.4 Economic Analysis: CAPEX, OPEX, and Payback Period

The following section presents the economic feasibility analysis of the proposed anaerobic digestion plant. To comprehensively evaluate the financial viability of the project, a detailed breakdown of both Capital Expenditures (CAPEX) and Operational Expenditures (OPEX) is required. CAPEX encompasses the initial investment costs associated with the engineering, procurement, and construction of the facility, including the reactors, mixing systems, and associated civil works. Conversely, OPEX accounts for the recurring annual expenses necessary to operate and maintain the plant, such as energy consumption, maintenance, and personnel costs.

The costs will be detailed and analyzed by breaking down the individual operational units and phases of the plant. Following the cost assessment, a final financial evaluation will be conducted to determine the Payback Period, thereby validating the overall economic sustainability of the investment.

4.4.1 Pre-treatment: Shredding Unit Costs

The primary mechanical pre-treatment phase involves the shredding of the incoming organic matter. This step is crucial to reduce the particle size, homogenize the substrate, and increase the specific surface area available for bacterial attack, thereby optimizing the anaerobic digestion kinetics.

The cost estimation for this unit was developed using a factored estimation method based on the primary Purchased Equipment Cost (PEC). The base cost of the industrial shredder was estimated at 120,000.00 €. The ancillary Capital Expenditures (CAPEX)—including installation, electrical panels, civil works (e.g., foundations and loading pits), conveyor belts/augers, and engineering—were calculated as conservative percentages of the PEC. This approach yields a total CAPEX of 168,000.00 €.

Regarding the Operational Expenditures (OPEX), two main parameters were evaluated: energy consumption and maintenance. The energy cost was calculated assuming a specific industrial electricity tariff of 0.20 €/kWh. Maintenance costs were estimated at 10% of the total CAPEX per year. This relatively high maintenance factor is fully justified by the heavy-duty nature of the shredding process, which induces significant mechanical wear and tear on the cutting blades and rotors. The breakdown of both CAPEX and OPEX is detailed in Table **??**.

Item	**Estimated Cost**	**U. M.**
Shredder (Base Cost)	120,000.00	€
Installation (10% of base)	12,000.00	€
Electrical Panels (5% of base)	6,000.00	€
Civil Works (10% of base)	12,000.00	€
Conveyors/Augers (10% of base)	12,000.00	€
Engineering (5% of base)	6,000.00	€
Total CAPEX	**168,000.00**	**€**

Table 4.9: Capital Expenditures (CAPEX) for the shredding unit

Item	**Estimated Cost**	**U. M.**
Maintenance (10% of total CAPEX)	16,800.00	€/year
Energy Consumption Cost	12,181.99	€/year
Total OPEX	**28,981.99**	**€/year**

Table 4.10: Annual Operational Expenditures (OPEX) for the shredding unit

4.4.2 Pre-treatment: Pre-mixing Unit Costs

Following the shredding phase, the organic substrate is transferred to a pre-mixing tank. In this unit, the biomass is homogenized and brought to the optimal total solids concentration required for the dry anaerobic digestion process. This is typically achieved through mechanical agitation and, if necessary, the recirculation of process liquid.

The Capital Expenditures (CAPEX) for this unit include the construction of the mixing tank, which must be built using highly durable, corrosion-resistant materials (such as coated reinforced concrete or stainless steel) due to the aggressive nature of the organic waste. Based on a specific construction cost of 1,800.00 €/m^3 and a required volume of 15 m^3, the tank cost was estimated at 27,000.00 €. Additional equipment costs include a 18.5 kW electric motor for the agitator, a high-density discharge pump, and the necessary control instrumentation, bringing the total CAPEX to 40,500.00 €.

The Operational Expenditures (OPEX) are primarily driven by the continuous energy demand of the mixer and the discharge pump, estimated at 79,200 kWh/year. Applying the industrial energy tariff of 0.20 €/kWh, the annual energy cost is 15,840.00 €. Furthermore, an annual maintenance cost was conservatively estimated at 3% of the total CAPEX to account for the wear and tear of the electromechanical components.

The detailed cost breakdowns for the pre-mixing unit are presented in Table 4.11 and Table 4.12.

Item	**Estimated Cost**	**U. M.**
Tank Construction (Volume: 15 m^3)	27,000.00	€
Electric Motor (18.5 kW)	6,000.00	€
Discharge Pump	5,000.00	€
Instrumentation & Control	2,500.00	€
Total CAPEX	**40,500.00**	**€**

Table 4.11: Capital Expenditures (CAPEX) for the pre-mixing unit

Item	**Estimated Cost**	**U. M.**
Energy Consumption Cost	15,840.00	€/year
Maintenance (3% of total CAPEX)	1,215.00	€/year
Total OPEX	**17,055.00**	**€/year**

Table 4.12: Annual Operational Expenditures (OPEX) for the pre-mixing unit

4.4.3 Anaerobic Digester Unit Costs

The core of the plant consists of a dual-reactor system ($n = 2$), utilizing single-phase dry *plug-flow* anaerobic digesters. The Capital Expenditures (CAPEX) for this unit primarily involve the civil engineering works, including the construction of the reinforced concrete tanks, the application of thermal insulation to maintain optimal process temperatures, and the installation of the gas-tight sealing systems to capture the biogas.

The investment cost was evaluated using a specific construction cost of 350.00 €/m^3. Applying this parameter to the total effective design volume of both reactors combined ($V_{tot} = 1357.16\ m^3$), the total estimated CAPEX for the digester structures amounts to 475,008.80 €.

Regarding the Operational Expenditures (OPEX), the two main components are the energy required for the plant operations (such as the mixing system and heating requirements) and the structural maintenance. Based on the industrial energy tariff of 0.20 €/kWh, the annual energy cost for the dual-reactor system was calculated at 18,162.56 €. The annual maintenance cost was estimated at 2% of the total CAPEX. This percentage is significantly lower than that of the mechanical pre-treatment units, reflecting the static nature of the main civil structures, yielding an annual maintenance cost of 9,500.18 €.

The detailed cost breakdowns for the complete anaerobic digestion unit are presented in Table 4.13 and Table 4.14.

Item	**Estimated Cost**	**U. M.**
Specific Construction Cost	350.00	€/m^3
Total Design Volume ($n = 2$)	1357.16	m^3
Total CAPEX	**475,008.80**	**€**

Table 4.13: Capital Expenditures (CAPEX) for the dual-reactor anaerobic digestion unit

Item	**Estimated Cost**	**U. M.**
Energy Consumption Cost	18,162.56	€/year
Maintenance (2% of total CAPEX)	9,500.18	€/year
Total OPEX	**27,662.72**	**€/year**

Table 4.14: Annual Operational Expenditures (OPEX) for the dual-reactor anaerobic digestion unit

4.5 Cash Flow Analysis and Payback Period

The economic feasibility and the ultimate Payback Period of the anaerobic digestion plant are fundamentally driven by the net annual cash flows. These cash flows are generated not only by the production of renewable energy but also by the significant reduction in waste management costs. By treating the organic fraction of the municipal solid waste (FORSU) generated on-site, the Gardaland resort establishes a circular economy model that provides substantial economic returns alongside its environmental benefits.

The plant is estimated to operate for 8,000 hours per year. The biogas produced is converted into electrical and thermal energy via a Combined Heat and Power (CHP) cogeneration unit. Based on an electricity tariff of 0.18 €/kWh, the annual electrical revenue is calculated at 186,390.72 €. Concurrently, the recovered thermal energy, valued at a tariff of 0.055 €/kWh, contributes an additional 22,655.41 € annually.

Furthermore, processing the organic waste internally eliminates the need for external landfill or treatment facility disposal. Assuming an avoided disposal cost of 5.00 €/t for the total annual mass of 5,076 tons, the facility realizes a direct saving of 25,380.00 € per year. The combination of direct energy revenues and avoided costs results in a total annual economic benefit of 234,426.13 €, as detailed in Table 4.15.

Item	Value	U. M.
Electricity Tariff	0.18	€/kWh
Operating Hours	8,000.00	h/year
Electrical Revenues	**186,390.72**	**€/year**
Thermal Tariff	0.055	€/kWh
Thermal Revenues	**22,655.41**	**€/year**
Avoided Disposal Cost	5.00	€/t
Disposal Savings	**25,380.00**	**€/year**
Total Annual Revenues	**234,426.13**	**€/year**

Table 4.15: Annual revenues and avoided costs generated by the plant

To proceed with the evaluation of the investment's profitability, it is necessary to consolidate the financial outputs from all the individual process units discussed in the previous sections. Table 4.16 provides the aggregated summary of the Capital Expenditures (CAPEX) and the Operational Expenditures (OPEX) for the shredder, the pre-mixer, and the anaerobic digester system. These aggregated values will serve as the baseline for calculating the net cash flows and the final Payback Period.

Expenditure	Shredder (€)	Pre-mixer (€)	Digester (€)	Total (€)
Total OPEX (Annual)	28,981.99	17,055.00	18,581.45	**64,618.44**
Total CAPEX	168,000.00	40,500.00	475,008.81	**683,508.81**

Table 4.16: Summary of total CAPEX and OPEX for the plant units

The final step in the economic assessment of the anaerobic digestion plant is the evaluation of the investment recovery time. To achieve this, it is first necessary to determine the Net Annual Cash Flow generated by the facility under nominal operating conditions. This value is derived by subtracting the aggregated annual Operational Expenditures (OPEX) from the Total Annual Revenues (which encompass both direct energy sales and avoided disposal costs).

As summarized in Table 4.17, the total annual OPEX amounts to 64,618.44 €, while the total revenues reach 234,426.13 €. This generates a robust nominal Net Cash Flow of 169,807.68 € per year against a total Capital Expenditure (CAPEX) of 683,508.81 €.

Item	Value	U. M.
Total Annual Revenues	234,426.13	€/year
Total OPEX	64,618.44	€/year
Net Annual Cash Flow	**169,807.68**	**€/year**
Total CAPEX	**683,508.81**	**€**
Nominal Break-even (Base Calculation)	4.03	years

Table 4.17: Nominal Net Cash Flow calculation

While the base nominal calculation suggests an initial break-even point of approximately 4.03 years, a comprehensive 30-year financial model was developed to provide a more rigorous risk assessment. The dynamics of investment recovery over the plant's entire lifespan are illustrated in Figure 4.4.

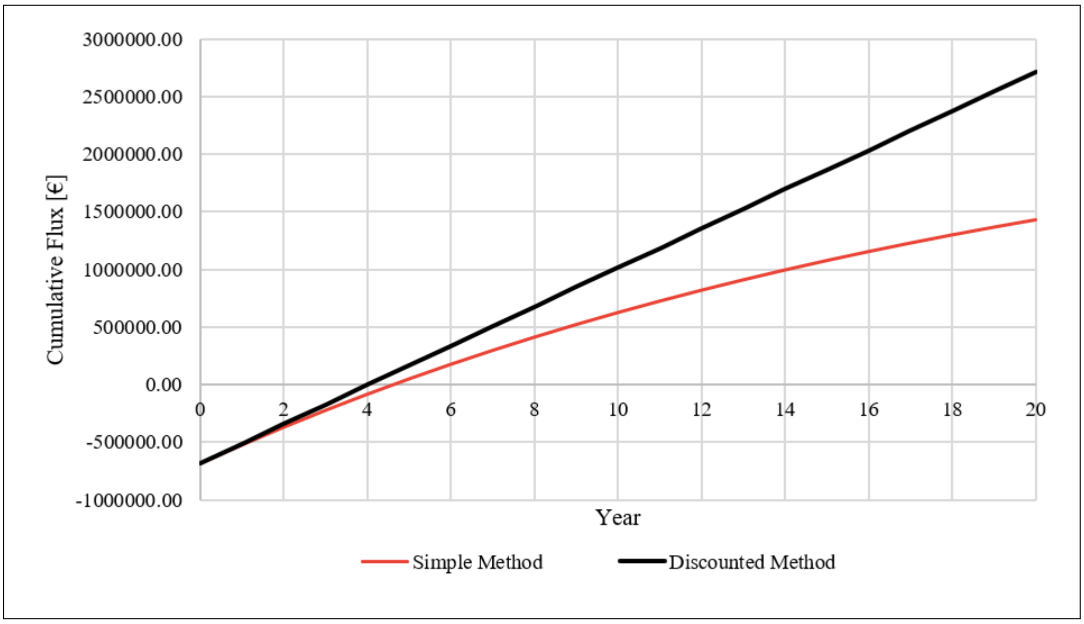

Figure 4.4: Cumulative cash flow over a 30-year horizon comparing Simple and Discounted Payback methodologies.

The chart compares the cumulative cash flows using two distinct financial methodologies. The steeper curve represents the Simple Payback Method, which evaluates the continuous accumulation of net revenues without considering the time value of money. Under this assumption, the project reaches its break-even point (where the cumulative cash flow crosses the zero axis) around 4 years.

Conversely, the lower curve represents the Discounted Payback Method. To provide a more conservative and realistic assessment, this methodology discounts future cash flows back to their present value by applying a Weighted Average Cost of Capital (WACC) of 5%. Because future revenues are effectively devalued by the cost of capital and inflation, the curve rises more gradually. Under this rigorous scenario, the investment is fully recovered in approximately 7.2 years.

Both metrics confirm the strong economic viability of the project. A discounted payback period of 5 years is highly competitive for a heavy infrastructural environmental engineering project, demonstrating that the facility will generate substantial and stable profit for the vast majority of its operational lifespan.

5 Possible Scenario Configurations

Having established the baseline technical sizing and economic feasibility of the anaerobic digestion plant, it is essential to explore alternative operational strategies to maximize the valorization of the produced biogas. The baseline financial model assumes a standard, nominal energy recovery profile; however, the inherent flexibility of a Combined Heat and Power (CHP) system allows for multiple utilization pathways for both the generated electrical and thermal energy.

The following subsections will systematically analyze various energy mix scenarios. These configurations will evaluate the technical and economic impacts of shifting the balance between on-site self-consumption—aimed at directly offsetting the resort's substantial energy demands—and the injection of surplus energy into the national grid. Each scenario will be assessed to determine its influence on overall thermodynamic efficiency, operational autonomy, and ultimate financial profitability. By comparing these alternative configurations against the baseline, it is possible to identify the most resilient and optimized operational framework, capable of adapting to fluctuating energy market tariffs and seasonal demand variations.

5.1 Scenario 1: 100% Solar Energy

After analysing the different renewable energy sources that could be applied to the Gardaland case study, different energy scenarios were developed and compared. The aim was to understand which combination of technologies could reduce the dependence on electricity from the grid and increase the use of renewable energy. Among the possible solutions, solar energy was considered one of the most suitable options, mainly because the site already has a photovoltaic installation and because the area presents good solar availability during the year.
In the first scenario, a solution based entirely on solar energy was proposed. The idea was to expand the photovoltaic plant already present at Gardaland, increasing the installed capacity until the annual electricity production could cover the total energy demand of the park. This scenario represents a 100% solar energy configuration, where the electricity required by the site is produced by photovoltaic panels. The choice of this solution is linked to the possibility of using existing surfaces and infrastructure, while also taking advantage of the high solar irradiance available during the operating season.
For the design of the photovoltaic system, the selected panel model was the SunPower Maxeon 3, with a nominal power of 400 W. This type of panel was chosen because it has good efficiency and is suitable for installations where a high energy production is required. The module efficiency considered in the design was 0.20, meaning that 20% of the solar radiation received by the panel is converted into electrical energy. In order to reach the required production, the system was designed with 29,968 photovoltaic panels. Considering the dimensions of the panels, the total surface area to be covered is approximately 53,043 m^2. This value is very important because it gives an idea of the physical feasibility of the project: the scenario requires a large available area, so the use of roofs, parking areas, service buildings or other suitable surfaces must be carefully evaluated.

In the following graph (fig. 5.1) the x-axis represents the operating hours, while the y-axis represents the cumulative electricity produced, expressed in kWh. The curve increases progressively because the graph does not show the instantaneous production, but the total energy produced from the beginning of the year up to each hour. The final value reached by the curve is about 23 GWh, which corresponds to the total annual production of the designed photovoltaic system.

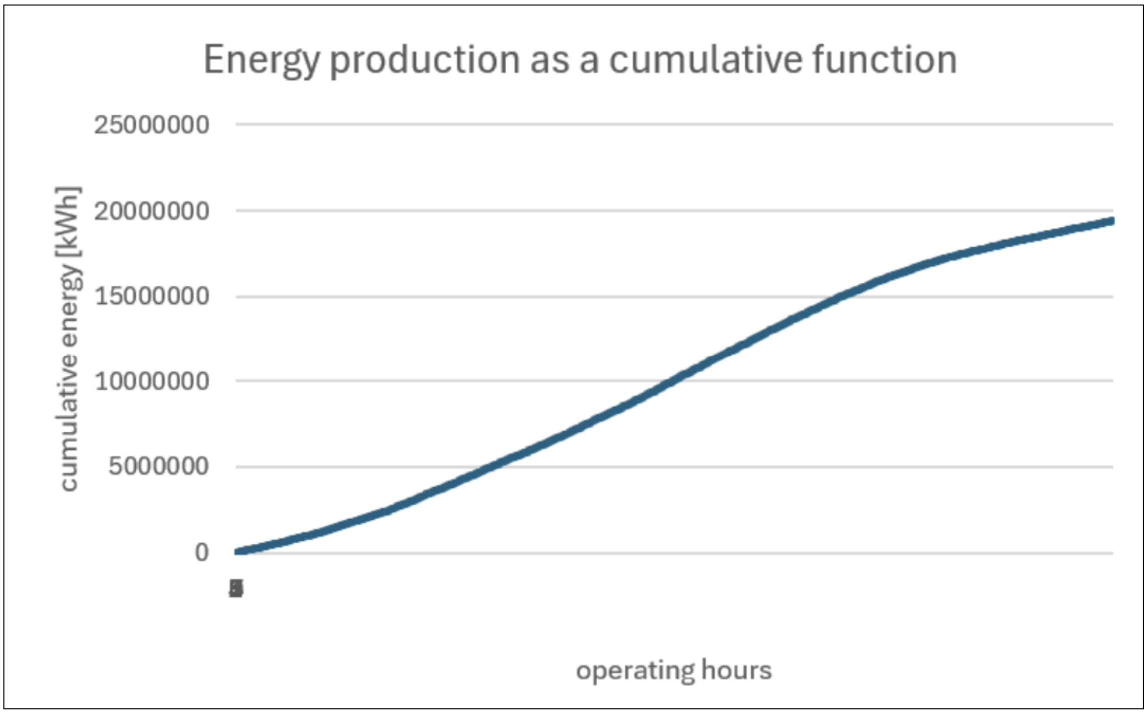

Figure 5.1: Cumulative energy production of the photovoltaic plant during a year

The shape of the curve depends on two main elements: the technical characteristics of the photovoltaic plant and the solar irradiance values used in the calculation. The panel efficiency, the number of modules and the total installed surface determine the maximum potential production of the system. However, the real hourly production also depends on the solar radiation available during the year. For this reason, production is not constant: it is higher in periods with stronger irradiance and lower during hours or seasons with less solar radiation. The cumulative curve therefore grows more rapidly when solar conditions are favourable and more slowly when irradiance is reduced.

However, in order to implement this scenario in a realistic way, the photovoltaic plant must be supported by a battery storage system. Even if the annual PV production is equal to the annual electricity consumption of Gardaland, equal to about 23 GWh/year, the production and the demand do not occur at the same time. Solar energy is produced mainly during daylight hours and depends on solar irradiance, while the park consumes electricity according to its own operating profile. For this reason, there are moments when the PV plant produces more energy than required and other moments when the production is not sufficient to cover the demand.
The storage system is therefore necessary to improve the self-consumption of the electricity produced by the photovoltaic plant. In this scenario, the battery system is composed of 3 batteries, each with a capacity of 5,000 kWh, for a total storage capacity of 15 MWh. According to the energy balance, about 4.56 GWh/year can be supplied by the battery, reducing the amount of electricity purchased from the grid. However, the system still requires some grid interaction: the total grid purchase is about 11.35 GWh/year, while the exported or curtailed energy is about 0.41 GWh/year. This shows that storage improves the performance of the solar scenario, but it does not completely eliminate the temporal mismatch between production and consumption.
From an economic point of view, the introduction of batteries represents an important additional cost. Considering a battery CAPEX of 250 €/kWh, the total investment cost for the storage system is about 3.75

million euros. The annual operation and maintenance cost is estimated as 2% of the battery CAPEX, corresponding to about 75,000 €/year. With an electricity purchase price of 0.18 €/kWh and a selling price of 0.08 €/kWh, the annual saving is about 455,784 €/year, while the net annual saving, after maintenance costs, is about 380,784 €/year. Therefore, the battery system makes the scenario more technically reliable, but it also has a significant impact on the total investment.

Overall, Scenario 1 shows that a full solar solution could theoretically produce enough renewable electricity to cover the annual energy demand of Gardaland. In this case, the large number of panels does not represent a major limitation in terms of available space, because Gardaland has a very large parking area that could be used for the installation of the photovoltaic system. Covering part of the parking area with solar panels could also have an additional advantage, because the panels could work as shading structures for cars while producing renewable electricity. For this reason, the scenario appears feasible from a spatial point of view, even if further technical and economic evaluations would still be necessary to define the final layout, the connection to the grid, and the investment cost. This first scenario therefore represents a strong renewable energy solution, based entirely on solar production and supported by the availability of a suitable installation area within the park.

5.2 Scenario 2: 100% Wind Energy

Scenario 2 evaluates the technical, logistical, and financial feasibility of expanding the existing wind farm infrastructure located at Rivoli Veronese (Monte Mesa). The proposal consists of installing four additional industrial-scale wind turbines of the LTW90 2000 model (2.0 MW each), for a total new capacity of 8.0 MW. Based on meteorological analysis and detailed energy modelling, this expansion is projected to generate 25.82 GWh/year, fully covering the resort's annual electricity demand and producing a net annual surplus of 2.82 GWh, which can be monetised through grid injection.

5.2.1 Wind turbines

The turbine model selected for the capacity expansion is the Leitwind LTW90 2000, a 2.0 MW horizontal-axis wind turbine optimised for medium-wind conditions. This unit features a 90-metre rotor diameter and a 90-metre steel monopole hub height, achieving an ideal balance between aerodynamic energy capture and structural loading constraints. The key operational parameters are summarised in the table below:

Table 5.1: Technical specifications of the Leitwind LTW90 2000 turbine.

Parameter	Value	Unit
Nominal rated capacity (per turbine)	2.0	MW
Hub height	90	m
Rotor diameter	90	m
Cut-in wind speed	4.0	m/s
Rated wind speed	9.0	m/s
Cut-out wind speed	25.0	m/s
Number of new turbines	4	–
Total new installed capacity	8.0	MW

The power curve of the LTW90 2000 transitions linearly from the cut-in speed of 4.0 m/s, reaching its rated output of 2.0 MW at 9.0 m/s, after which blade pitch control maintains constant nominal power until the cut-out speed of 25.0 m/s, where the braking system engages. The combination of the wind speed distribution at Rivoli Veronese and this power curve is shown in Figure 5.2, which illustrates that the turbine operates within its productive range for a significant fraction of annual hours.

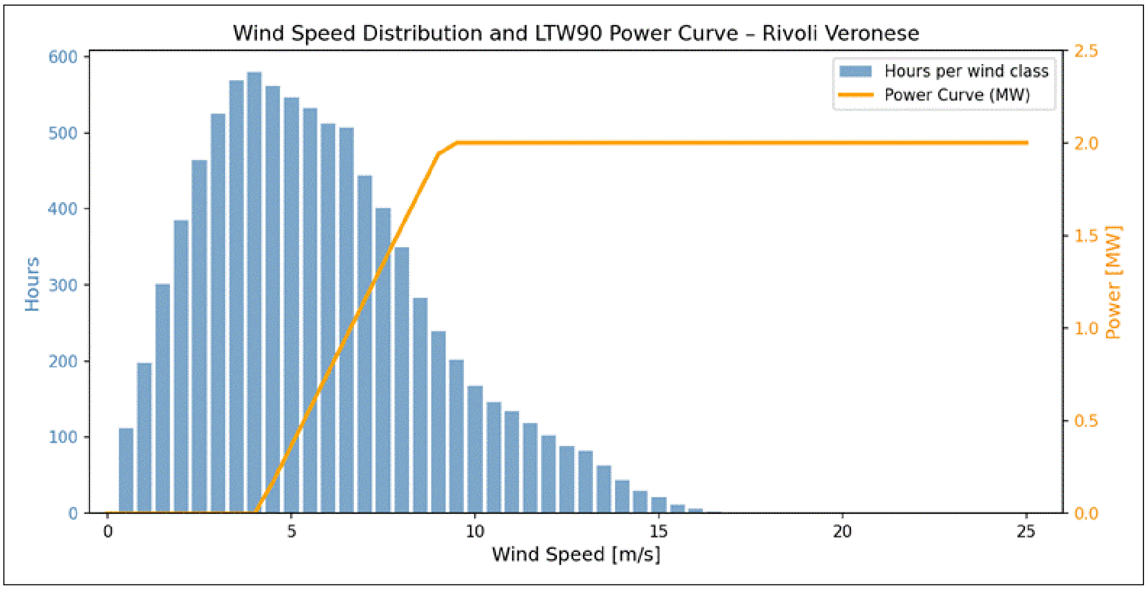

Figure 5.2: Wind speed distribution (hourly occurrences, blue bars) overlaid with the LTW90 2000 Power Curve (orange line) – Rivoli Veronese. The turbine operates in its productive range for the vast majority of annual hours, with rated power reached at approximately 9 m/s.

The Annual Energy Production (AEP) was calculated by mapping each hourly wind speed observation, corrected to hub height, against the turbine's power curve, and multiplying the resulting power output by four (for the four new turbines):

$$AEP_4 = 4 \cdot \Sigma \left[P(V_i) \cdot \Delta t \right] \qquad \text{for all } i \text{ hours in the dataset} \tag{5.1}$$

The resulting production figures are:

Table 5.2: Energy balance for 4 × LTW90 2000, 8 MW total.

Metric	Value
Annual energy production per turbine	6.45 GWh/year
Total AEP for 4 turbines	25.82 GWh/year
Gardaland total annual demand	23.00 GWh/year
Energy autonomy ratio (wind)	112.2%
Net annual energy surplus	2.82 GWh/year
Avoided CO_2 emissions (at 450 gCO_2/kWh)	11,619 tCO_2/year

Since the wind energy production asset in Rivoli Veronese is physically located approximately 15–18 km from Gardaland's electricity consumption points in Castelnuovo del Garda, a dedicated private cable connection would be cost-prohibitive and legally unfeasible, given the multiple municipalities and infrastructure corridors involved. The legal and logistical integration of the new wind capacity into Gardaland's energy supply is therefore structured through an off-site virtual Power Purchase Agreement (PPA), a well-established commercial and regulatory mechanism in the Italian and broader European energy markets.

Under this framework, the 8.0 MW wind array in Rivoli Veronese injects 100% of its generated electricity directly into the local high-voltage substation grid at Rivoli Veronese. This energy is then wheeled through the national transmission grid and the distribution network. Simultaneously, Gardaland Resort continues to draw electricity from its standard medium-voltage connection points in Castelnuovo del Garda. A tri-party balancing agreement is established between Gardaland (as buyer/offtaker), the wind asset special purpose vehicle (SPV, as seller/generator), and the Italian energy market manager (GME, *Gestore dei Mercati Energetici*). The physical grid acts as a virtual battery and transport mechanism, entirely eliminating the need for capital-intensive on-site battery energy storage systems.

The PPA mechanism is fully compliant with current Italian energy regulation and with the EU renewable energy directive, which explicitly promotes corporate PPAs as a tool to accelerate the energy transition. In Italy, off-site virtual PPAs are further facilitated by Decree 199/2021 and the evolving framework of Energy Communities (CER, *Comunità Energetiche Rinnovabili*) under Italian Law 199/2021.

A critical element of the PPA structure is the transfer of Guarantees of Origin (GO). Under European Directive 2018/2001/EU, for every megawatt-hour injected into the grid by the Rivoli Veronese wind turbines, a certified GO is electronically generated and registered in the national GO registry, managed by GSE (*Gestore dei Servizi Energetici*). These GOs are contractually transferred to Gardaland's account as part of the PPA agreement.

This mechanism provides full legal and commercial traceability: Gardaland can demonstrate to regulatory bodies, sustainability rating agencies, and its parent company Merlin Entertainments that 100% of its consumed electricity is matched by actual renewable wind generation. This enables Gardaland to legitimately claim "100% Wind Powered" status, in full alignment with Merlin Entertainments' published 2030 Net-Zero targets.

The PPA is therefore not merely a financial hedging instrument, it is the legal backbone of Gardaland's renewable energy certification, making it both technically sound and commercially robust.

The PPA framework offers a set of structural advantages beyond simple cost savings:

Table 5.3: Strategic benefits of the Off-Site PPA framework for Gardaland.

Strategic Benefit	Description
Price hedging	Gardaland locks in a fixed energy cost of € 50–60/MWh for 15–20 years, compared to the current industrial grid tariff of € 120–160/MWh in Italy. This provides protection against future energy market volatility.
24/7 Operations	Unlike solar-only solutions, wind production better complements a continuous demand profile, supporting hotel automation, climate control, and aquarium life-support systems throughout the day and year.
No BESS required	The grid acts as a virtual storage and balancing mechanism, avoiding the need for capital-intensive battery energy storage systems (BESS), thus significantly reducing CAPEX.
Surplus monetisation	The annual surplus of 2.82 GWh can be sold back to the grid or shared through a Renewable Energy Community (CER), generating additional revenue streams and CSR value.
Regulatory compliance	Full compliance with Italian Legislative Decree 199/2021 and related regulations, with streamlined permitting due to the site's existing industrial wind energy context.

5.2.2 Financial analysis

The total initial capital investment for the development, procurement, and commissioning of the 4-turbine, 8.0 MW expansion is estimated based on the installation cost of €1.5 million per MW of installed capacity, a conservative and industry-standard figure for Italian onshore wind projects of this scale.

Table 5.4: Capital expenditure breakdown for the 8 MW wind expansion.

CAPEX Component	Cost
Turbine installation (4 × 2 MW at €1.5M/MW)	€12.0 million
Total CAPEX	**€12.0 million**

The total CAPEX of €12.0 million encompasses turbine procurement and logistics (monopoles, nacelles, rotors), civil works (reinforced concrete pile foundations, crane pads), substation step-up transformers and grid connection fees, environmental impact assessment, local permitting, and engineering design.

Annual fixed operational costs are budgeted at €0.5 million per year. This covers scheduled preventive maintenance contracts, such as blade inspections, gearbox lubrication and control system monitoring, all-risk insurance and business interruption coverage, land lease royalties to landowners at Monte Mesa and administration and reporting costs for the PPA agreement. The OPEX estimate represents approximately 4.2% of CAPEX per year, which is consistent with industry benchmarks for onshore wind farms of this size.

Revenue is generated primarily through the avoided cost of purchasing grid electricity. The resort currently procures electricity at an industrial grid tariff estimated at €100/MWh. The wind production effectively displaces grid purchases at this rate. The 25.82 GWh/year of wind production therefore generates an annual avoided-cost revenue of:

$$\text{Annual revenue} = 25.82\ \text{GWh} \times €\,100/\text{MWh} = €\,2.582\ \text{million/year} \tag{5.2}$$

Additionally, the net annual surplus of approximately 2.82 GWh can be sold to the grid, at current GME zonal prices. This surplus revenue stream, although secondary, further improves the project's financial performance.

The net annual cash flow after OPEX is calculated as follows:

$$\text{Net annual cash flow} = \text{Revenue} - \text{OPEX} = €2.582\text{M} - €0.5\text{M} = €2.082 \text{ million/year} \tag{5.3}$$

The simple payback period is therefore:

$$PBP = \frac{CAPEX}{\text{Net annual cash flow}} = \frac{€12.0\text{M}}{€2.082\text{M}} \approx 5.42 \text{ years} \tag{5.4}$$

A simple payback period of approximately 5.42 years, for an asset with an engineering lifespan of 20–25 years, represents a highly attractive financial profile. Over the full 25-year operational life of the turbines, the project is projected to generate a cumulative net cash flow of approximately €40 million above the initial investment.

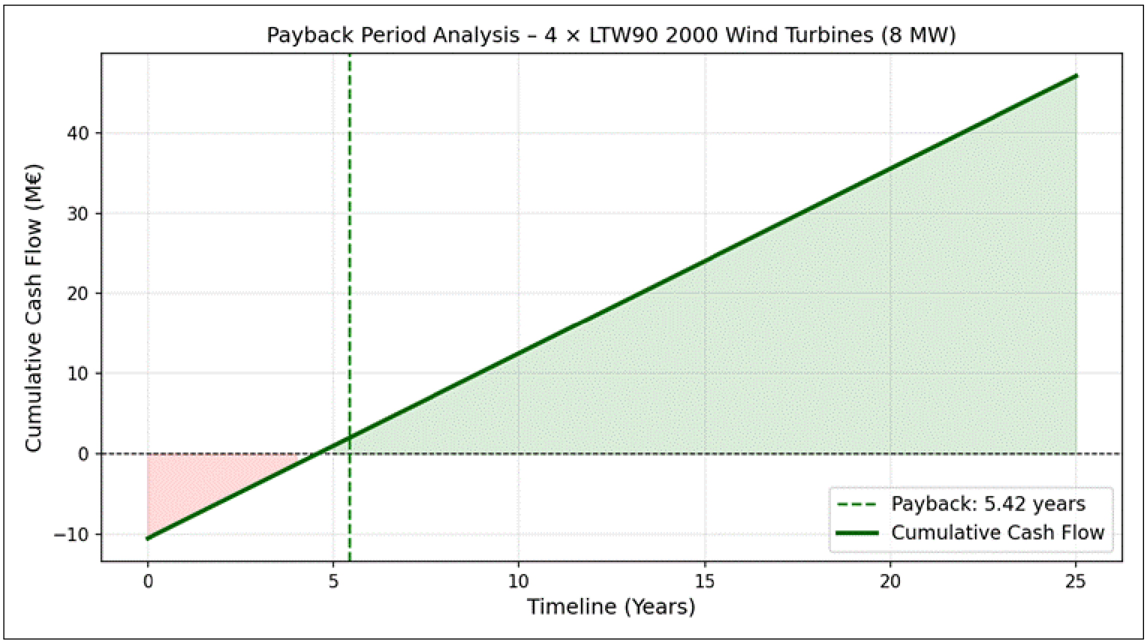

Figure 5.3: Cumulative Cash Flow Analysis: 4 × LTW90 2000 Wind Turbines. The project reaches simple payback at 5.42 years. Over a 25-year operational horizon, the cumulative net return far exceeds the initial capital investment.

Table 5.5: Summary of financial metrics for Scenario 2 wind energy expansion.

Financial Metric	Value
Total CAPEX	€12.0 million
Annual OPEX (fixed costs)	€0.50 million/year
Annual revenue (avoided grid cost + surplus)	€2.58 million/year
Net annual cash flow	€2.08 million/year
Simple payback period	5.42 years
Project lifespan	20–25 years
LCOE (Levelized Cost Of Energy)	$\sim$€50/MWh

The Levelized Cost Of Energy (LCOE) for this project is approximately €50/MWh, compared to the current Italian industrial grid tariff of €100–160/MWh. This price differential of €50–110/MWh represents Gardaland's structural energy cost advantage for every MWh produced over the project's lifetime. The LCOE figure also makes the project highly competitive with any alternative energy supply scenario and confirms that the wind expansion is the economically dominant option for long-term energy cost management.

5.2.3 Environmental impact

The most immediate environmental impact of this scenario is the near-complete decarbonisation of Gardaland resort's electricity consumption. Based on the Italian national grid average emission factor of 450 gCO_2/kWh (source: ISPRA national inventory data), the 25.82 GWh of wind energy produced annually will avoid approximately 11,619 tonnes of CO_2 equivalent per year. Over the 25-year lifetime of the project, this corresponds to total avoided emissions of approximately 290,000 tonnes of CO_2.

From a regulatory standpoint, the project benefits from a favourable baseline. The site at Monte Mesa, Rivoli Veronese, is already classified as an industrial wind zone under the Veneto region's spatial planning instruments. The presence of the four existing REpower MM92 turbines demonstrates that the site has already undergone the required Environmental Impact Assessment process and has community acceptance. The new installation of four additional turbines of a similar or slightly larger class within the existing wind farm perimeter significantly reduces the scope and duration of new regulatory approvals.

Mitigation measures to satisfy updated EIA requirements include: implementation of AI-based bird detection radar systems capable of triggering automatic turbine shutdown during migratory bird passage events, anti-reflective blade coatings to reduce visual impact and glint, and noise analysis to confirm compliance with Italian DPCM 14/11/97 acoustic limits at the nearest residential receptors.

5.3 Scenario 3: Integrated Mix

The integration of the biomass digester within the broader energy strategy is formalized in Scenario 3, designated as the "Integrated Mix." The overarching objective of this scenario is to achieve complete energy independence from the national grid by optimizing local resources and maximizing the use of existing infrastructure. This resilient and sustainable energy ecosystem relies on a synergistic combination of three sources. Specifically, the newly proposed anaerobic digester provides a crucial, continuous baseload power output, satisfying 4.5% of the total energy demand. The primary energy load is met by the implementation of wind energy, which accounts for the vast majority of the park's demand at 70.5%. The remaining 25.0% is supplied by solar energy, effectively leveraging the pre-existing photovoltaic parking canopies.

Given that the solar infrastructure is already operational, its associated installation cost is classified as a sunk cost and is strictly excluded from the new capital expenditure calculations. The total initial investment (I_{tot}) is exclusively defined by the sum of the capital expenditures required for the wind infrastructure and the anaerobic digester ($CAPEX_{wind} + CAPEX_{biomass}$). For each operational year, the aggregate net cash flow ($CF(t)$) is computed by subtracting the respective Operation and Maintenance (O&M) costs from the economic benefits generated by these two new assets. These annual cash flows are then discounted using a designated reference discount rate (r) and accumulated over time. The Payback Period corresponds to the year, n, in which the cumulative discounted net cash flows equal or exceed the total initial capital investment, satisfying the following condition:

$$\sum_{t=1}^{n} \frac{CF(t)}{(1+r)^t} \geq I_{tot} \tag{5.5}$$

By applying this financial framework to the technical sizing of the proposed energy mix, the overall investment for Scenario 3 yields a payback period of 14 years.

In conclusion, a discounted payback period of 14 years is not an optimal metric from a strictly short-term financial perspective. However, this extended timeframe must be evaluated within its broader strategic context. The significant upfront capital expenditure is justified by the creation of a highly resilient, self-sufficient energy ecosystem. By completely eliminating reliance on the national grid, the project shields the facility from future energy market volatility and price spikes. Therefore, the 14-year return should be viewed not merely as a delayed financial yield, but as a necessary structural investment to ensure long-term infrastructural stability, sustainability, and complete energy independence.

5.4 Scenario 4: Solar Energy and Wind Energy

After analysing the possibility of using only one renewable energy source, a further scenario was developed by combining solar energy and wind energy. This solution can represent a valid alternative because these two sources have different production profiles during the day and during the year. Solar panels produce energy mainly during daylight hours and their production strongly depends on solar irradiance. For this reason, the highest production is generally concentrated in the central hours of the day and in the sunnier months. Wind energy, instead, is less directly connected to the presence of sunlight and can also produce electricity during the night or in periods when solar production is lower. Therefore, the combination of these two technologies makes the energy system more balanced and more flexible.

In this scenario, the energy supply is divided between solar and wind energyt. The contribution of the two sources is assumed to be 50% from solar energy and 50% from wind energy. This balanced configuration has the objective of reducing the limits of a system based only on photovoltaic panels. In fact, a solar-only system can require a very large installed area and can produce a high surplus during the day, while still needing support during the night. By adding wind turbines, part of the demand can be covered in hours when photovoltaic production is not available. This can reduce the mismatch between production and consumption and can improve the overall performance of the renewable energy system.

From an economic point of view, this hybrid scenario can also be considered interesting. Even if the installation of wind turbines introduces an additional investment cost, the combination of solar and wind energy can reduce the need for an oversized photovoltaic plant and can improve the use of the electricity produced. A more constant renewable production can also reduce the amount of energy that has to be bought from the grid. For this reason, the solar-wind configuration can be a valid alternative not only from a technical point of view, but also from an economic point of view, especially when the objective is to cover a very high

percentage of the annual electricity demand with renewable energy.
However, batteries are still included in this scenario. Their role remains important because renewable energy production is variable and does not always occur at the same time as energy consumption. The batteries allow the system to store the excess electricity produced during periods of high solar or wind generation and to use it later, when production is lower than demand. This improves self-consumption and reduces the dependence on the external grid. At the same time, batteries represent one of the most expensive components of the system. Their high cost has a significant impact on the total investment and must be carefully considered in the economic evaluation. For this reason, even if storage improves the technical performance of the system, it also increases the initial cost of the project.
The next graph (fig. 5.4) compares the energy production from the combined solar and wind system with the energy consumption of Gardaland during the year. As can be observed, the renewable production is variable, because it depends on solar irradiance and wind availability. The consumption profile also changes during the year, with higher values in the period when the park has greater activity and higher energy needs. Despite these variations, the total energy produced by the hybrid renewable system is able to compensate the total energy consumption of Gardaland on an annual basis. This means that, considering the overall energy balance, the combination of photovoltaic panels and wind turbines can generate enough electricity to cover the annual demand of the park.

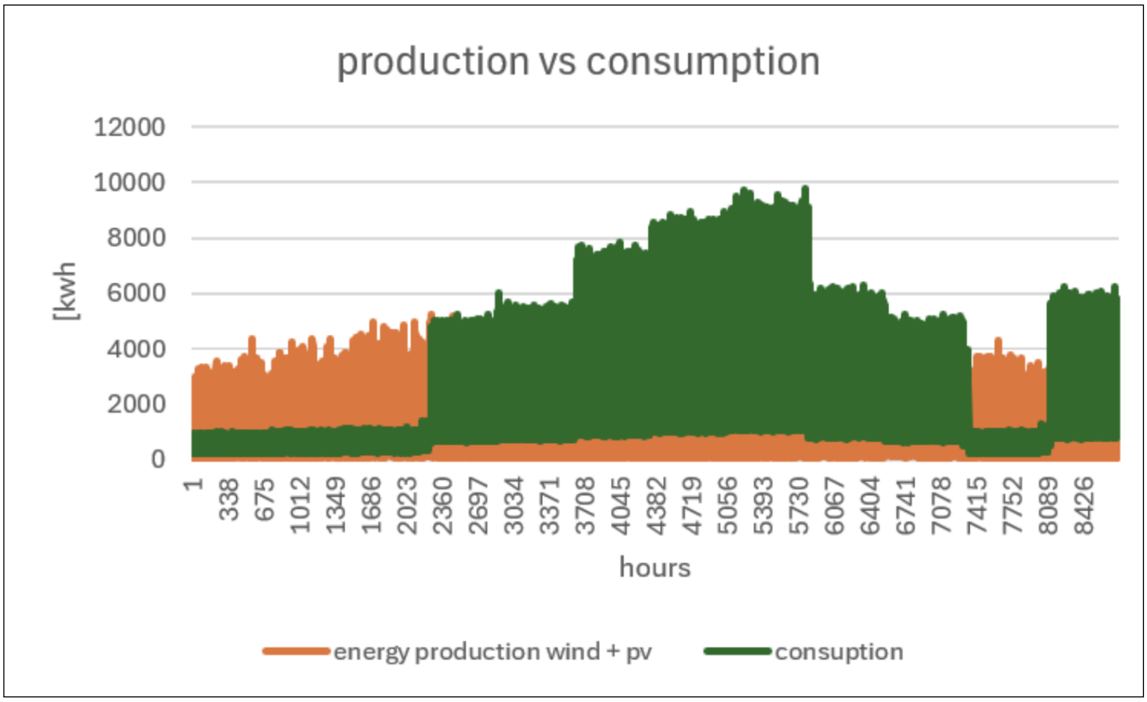

Figure 5.4: Energy production from the combined solar and wind system with the energy consumption of Gardaland during the year

In conclusion, Scenario 4 shows that a mixed renewable system based on solar and wind energy can be a strong solution for Gardaland. The two sources complement each other because they do not produce electricity in the same way and at the same time. This makes the system more reliable than a solution based on only one renewable source. Nevertheless, the presence of batteries is still necessary to manage the differences between production and consumption, and their high cost remains a critical aspect. Overall, this scenario represents a technically valid and potentially economically convenient option for increasing the renewable energy share and reducing the dependence on conventional energy sources

5.5 Comparison with other scenarios

Among all scenarios evaluated for Gardaland's renewable energy transition, Scenario 2 (100% wind) achieves the shortest simple payback period of 5.42 years, compared to 8.12 years for Scenario 1 (100% solar), 12.62 years for Scenario 3 (integrated mix with biomass), and 6.77 years for Scenario 4 (50% wind + 50% solar). The biomass component in Scenario 3 significantly increases the payback period and does not justify the added complexity and capital cost. Scenario 2 is therefore the financially dominant option and, combined with the existing photovoltaic canopy (Scenario 4 elements), represents the most balanced long-term strategy.

A key advantage of the pure wind approach is operational simplicity: a single technology, a single supplier relationship, a single PPA contract, and a single grid connection point at Rivoli Veronese. This reduces administrative complexity, counterparty risk, and lifecycle management overhead compared to a multi-technology hybrid scenario.

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